

Lecture 13: Data Visualization Case Study

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October 14, 2025

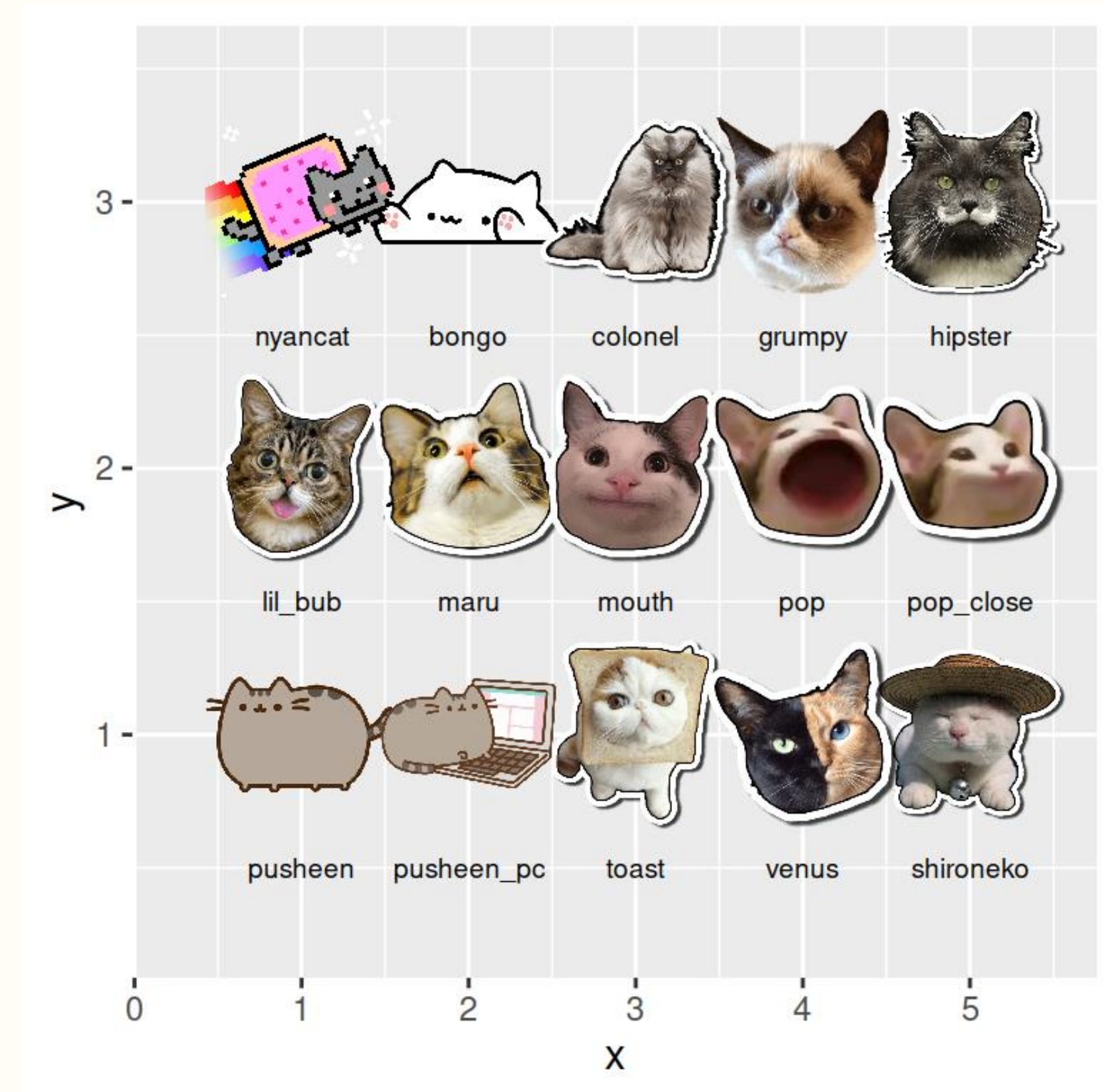
Recipe of the Day!

Spaghetti Carbonara



Announcements

- Homework #3 due Friday, October 24 by 11:59pm
- Lab will be an extended office hour this week
- Mid-course evaluation is live!



ggcats and geom_cat

Coding question of the day

Using the `murders` dataset from the `dslabs` package and the `state.x77` dataset that is built in to R, make a scatterplot with the total number of gun murders (`total`) on the y-axis and land area in square miles (`Area`) on the x-axis. You will need to use the `left_join` function to join the `state.x77` dataset to the `murders` dataset using the `state` column.

Bonus challenge: are there any NAs in the merged dataset? How are they created?

Be sure to run this code first

```
# Load necessary libraries
library(dplyr)
library(dslabs)
library(ggplot2)

# Load the state.x77 dataset and convert to a data frame
data("state")
state_data <- as.data.frame(state.x77)

# Add the state names to the state.x77 dataset
state_data$state <- rownames(state_data)

head(state_data)

# Load the murders dataset from the dslabs package
data("murders")
```

Comparing Distributions

Comparing Distributions Roadmap

1. Reordering
2. Smooth density plots

Comparing Distributions

Reordering

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Gapminder regions

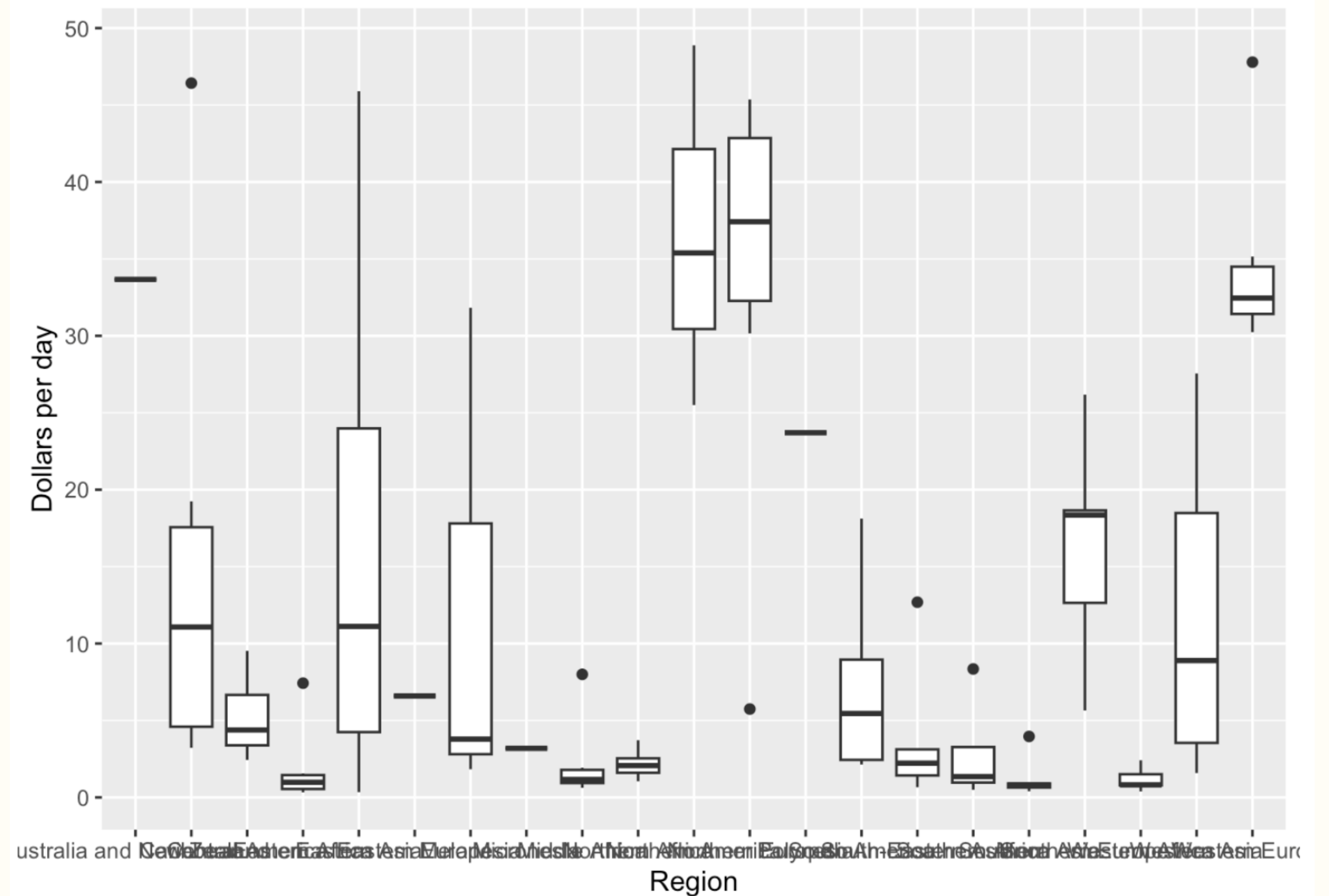
```
levels(gapminder$region)
```

```
## [1] "Australia and New Zealand" "Caribbean"
## [3] "Central America"          "Central Asia"
## [5] "Eastern Africa"           "Eastern Asia"
## [7] "Eastern Europe"           "Melanesia"
## [9] "Micronesia"               "Middle Africa"
## [11] "Northern Africa"          "Northern America"
## [13] "Northern Europe"          "Polynesia"
## [15] "South America"            "South-Eastern Asia"
## [17] "Southern Africa"          "Southern Asia"
## [19] "Southern Europe"          "Western Africa"
## [21] "Western Asia"             "Western Europe"
```

```
length(levels(gapminder$region))
```

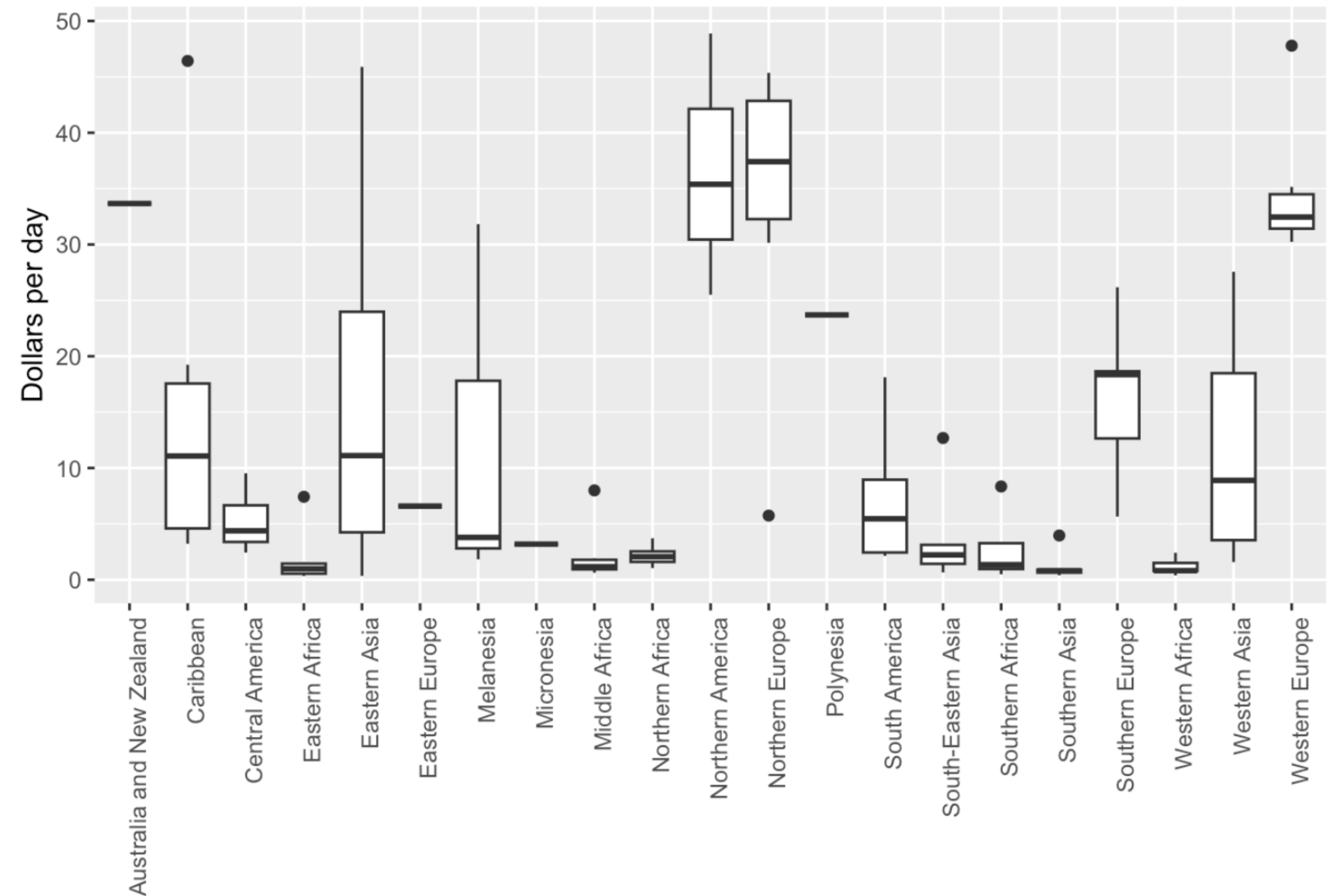
```
## [1] 22
```

```
gapminder %>%
  filter(year == past_year & !is.na(gdp)) %>%
  ggplot(aes(region, dollars_per_day)) +
  geom_boxplot() +
  xlab("Region") +
  ylab("Dollars per day")
```



Vertical axis labels

```
gapminder %>%  
  filter(year == past_year & !is.na(gdp)) %>%  
  ggplot(aes(region, dollars_per_day)) +  
  geom_boxplot() +  
  theme(axis.text.x = element_text(angle = 90, hjust = 1)) +  
  xlab("") +  
  ylab("Dollars per day")
```



reorder

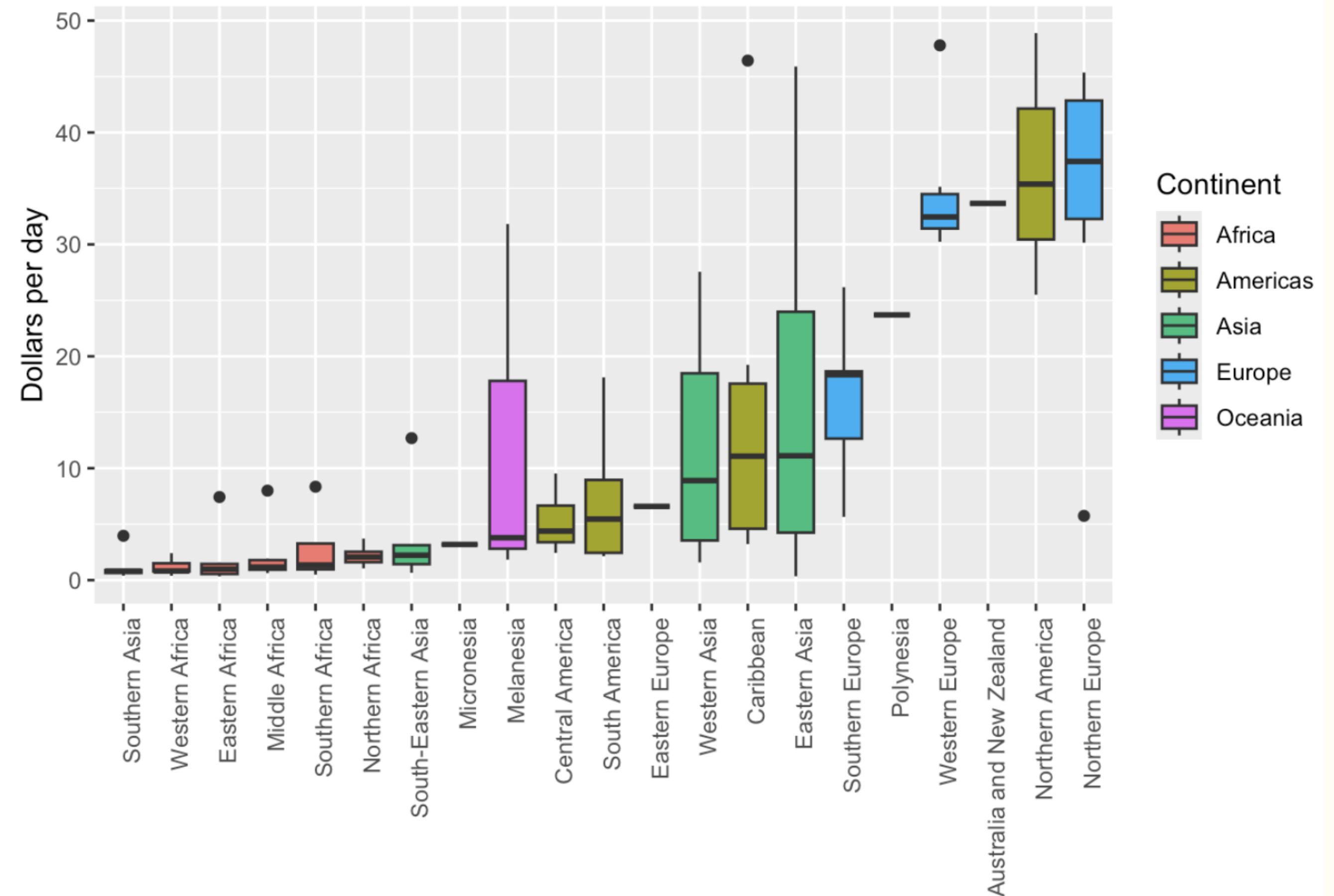
```
fac <- factor(c("Asia", "Asia", "West", "West", "West"))  
levels(fac)
```

```
## [1] "Asia" "West"
```

```
value <- c(10, 11, 12, 6, 4)  
fac <- reorder(fac, value, FUN = mean)  
levels(fac)
```

```
## [1] "West" "Asia"
```

```
gapminder %>%  
  filter(year == past_year & !is.na(gdp)) %>%  
  mutate(region = reorder(region, dollars_per_day, FUN = median)) %>%  
  ggplot(aes(region, dollars_per_day, fill = continent)) +  
  geom_boxplot() +  
  theme(axis.text.x = element_text(angle = 90, hjust = 1)) +  
  xlab("") +  
  ylab("Dollars per day") +  
  labs(fill = "Continent")
```



reorder

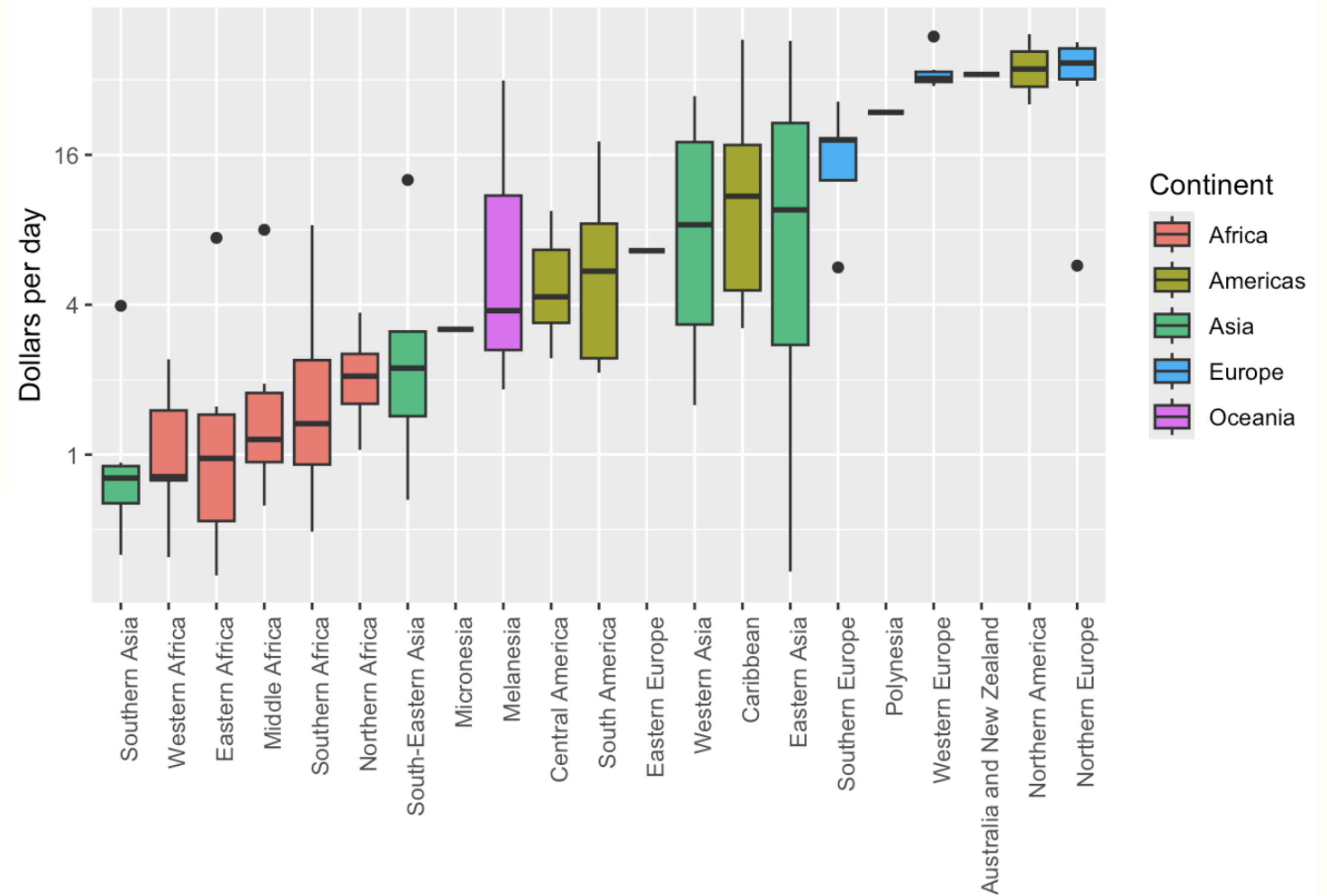
```
fac <- factor(c("Asia", "Asia", "West", "West", "West"))
levels(fac)
```

```
## [1] "Asia" "West"
```

```
value <- c(10, 11, 12, 6, 4)
fac <- reorder(fac, value, FUN = mean)
levels(fac)
```

```
## [1] "West" "Asia"
```

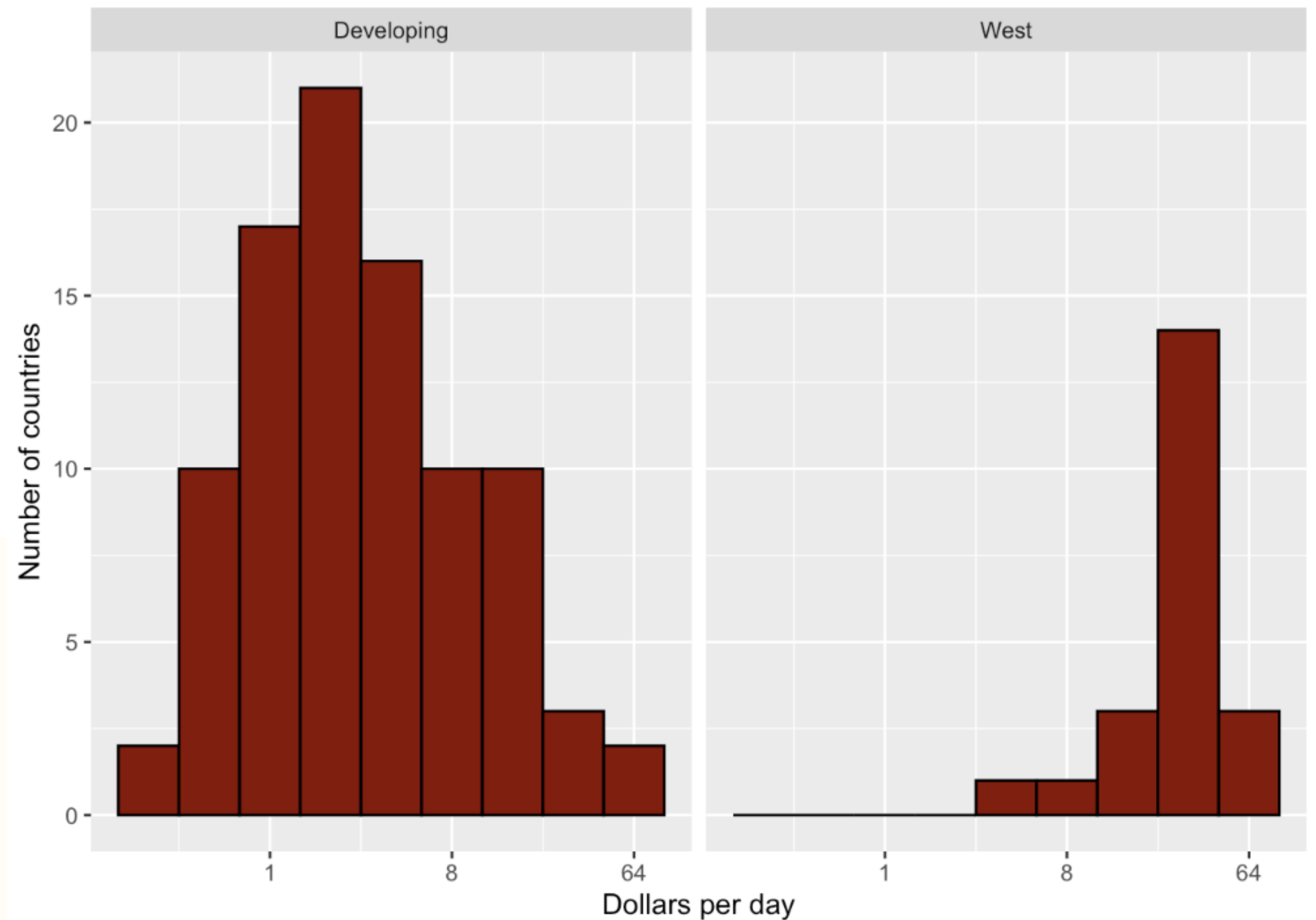
```
gapminder %>%
  filter(year == past_year & !is.na(gdp)) %>%
  mutate(region = reorder(region, dollars_per_day, FUN = median)) %>%
  ggplot(aes(region, dollars_per_day, fill = continent)) +
  geom_boxplot() +
  theme(axis.text.x = element_text(angle = 90, hjust = 1)) +
  xlab("") +
  ylab("Dollars per day") +
  labs(fill = "Continent") +
  scale_y_continuous(trans = "log2")
```



Compare across years

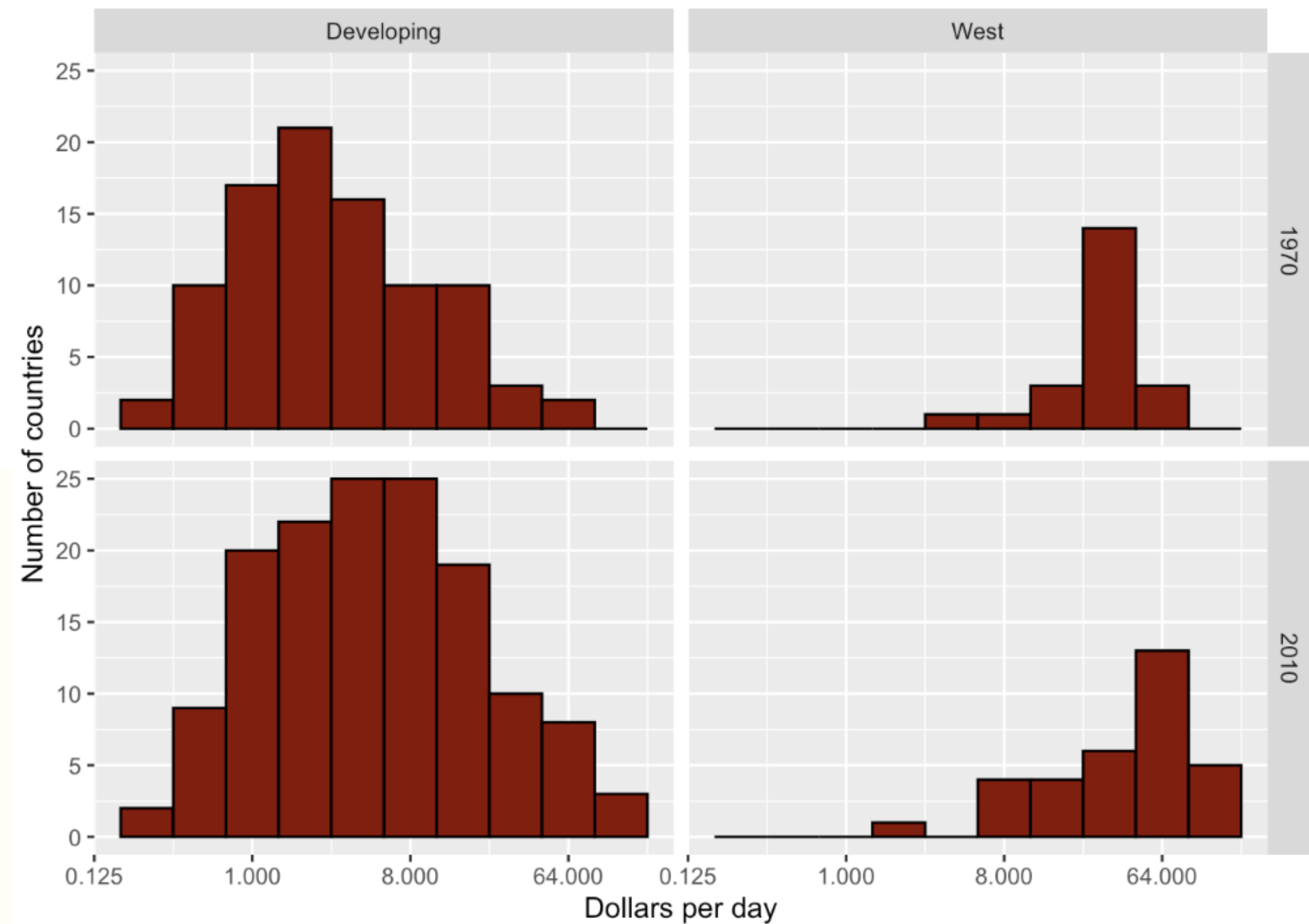
```
west <- c("Western Europe", "Northern Europe", "Southern Europe",  
         "Northern America", "Australia and New Zealand")
```

```
gapminder %>%  
  filter(year == past_year & !is.na(gdp)) %>%  
  mutate(group = ifelse(region %in% west, "West", "Developing")) %>%  
  ggplot(aes(dollars_per_day)) +  
  geom_histogram(binwidth = 1, color = "black") +  
  scale_x_continuous(trans = "log2") +  
  facet_grid(. ~ group) +  
  xlab("Number of countries") +  
  ylab("Dollars per day")
```



Facet by two variables

```
past_year <- 1970
present_year <- 2010
gapminder %>%
  filter(year %in% c(past_year, present_year) & !is.na(gdp)) %>%
  mutate(group = ifelse(region %in% west, "West", "Developing")) %>%
  ggplot(aes(dollars_per_day)) +
  geom_histogram(binwidth = 1, color = "black") +
  scale_x_continuous(trans = "log2") +
  facet_grid(year ~ group) +
  xlab("Number of countries") +
  ylab("Dollars per day")
```



Make valid comparisons

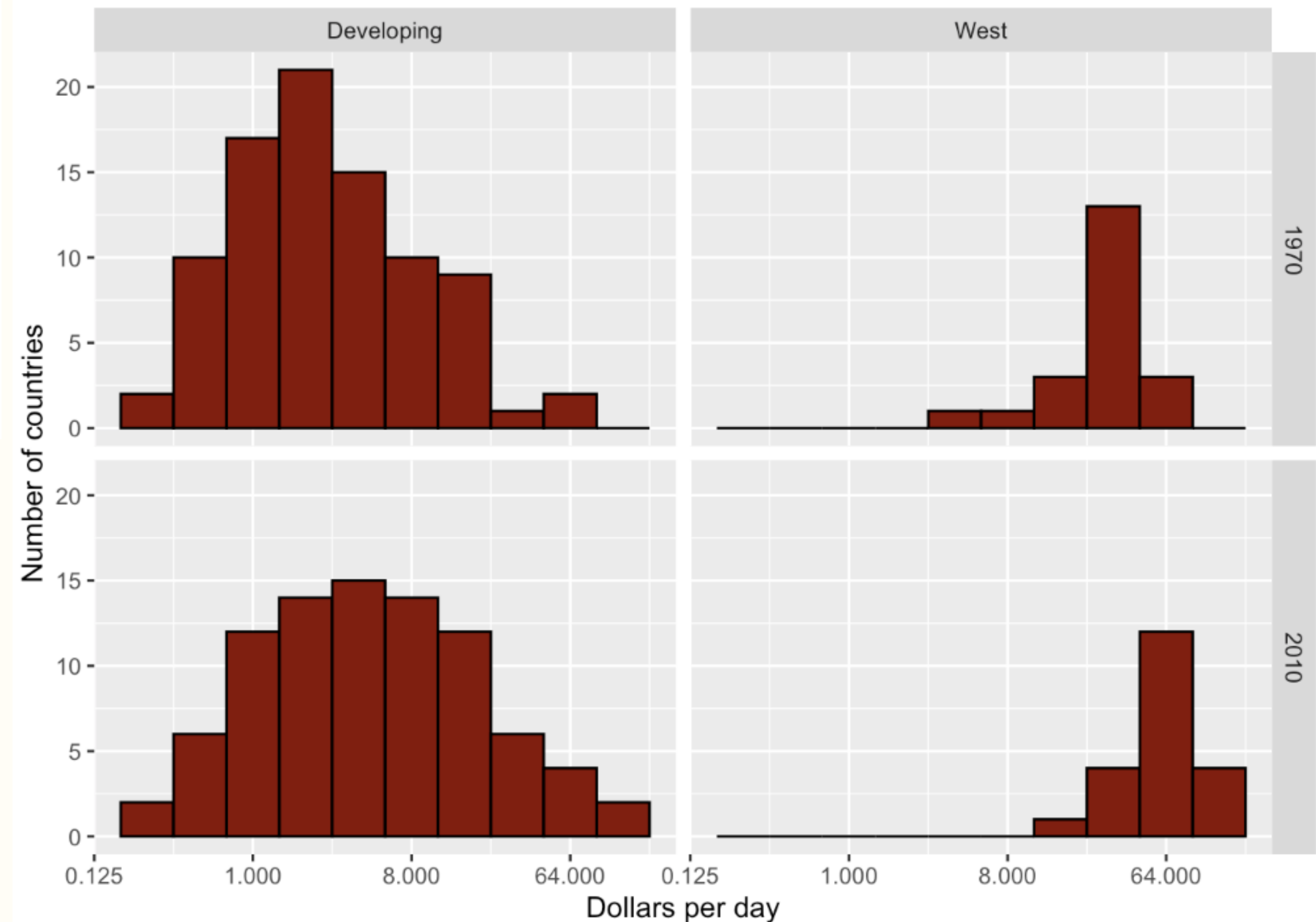
The number of countries represented in 1970 is less than the number in 2010

- More data for 2010
- Some countries split into multiple after 1970
- To make a **valid comparison** across years, we must have the **same countries for both years**

```
country_list_1 <- gapminder %>%  
  filter(year == past_year & !is.na(dollars_per_day)) %>%  
  .$country  
  
country_list_2 <- gapminder %>%  
  filter(year == present_year & !is.na(dollars_per_day)) %>%  
  .$country  
  
country_list <- intersect(country_list_1, country_list_2)
```

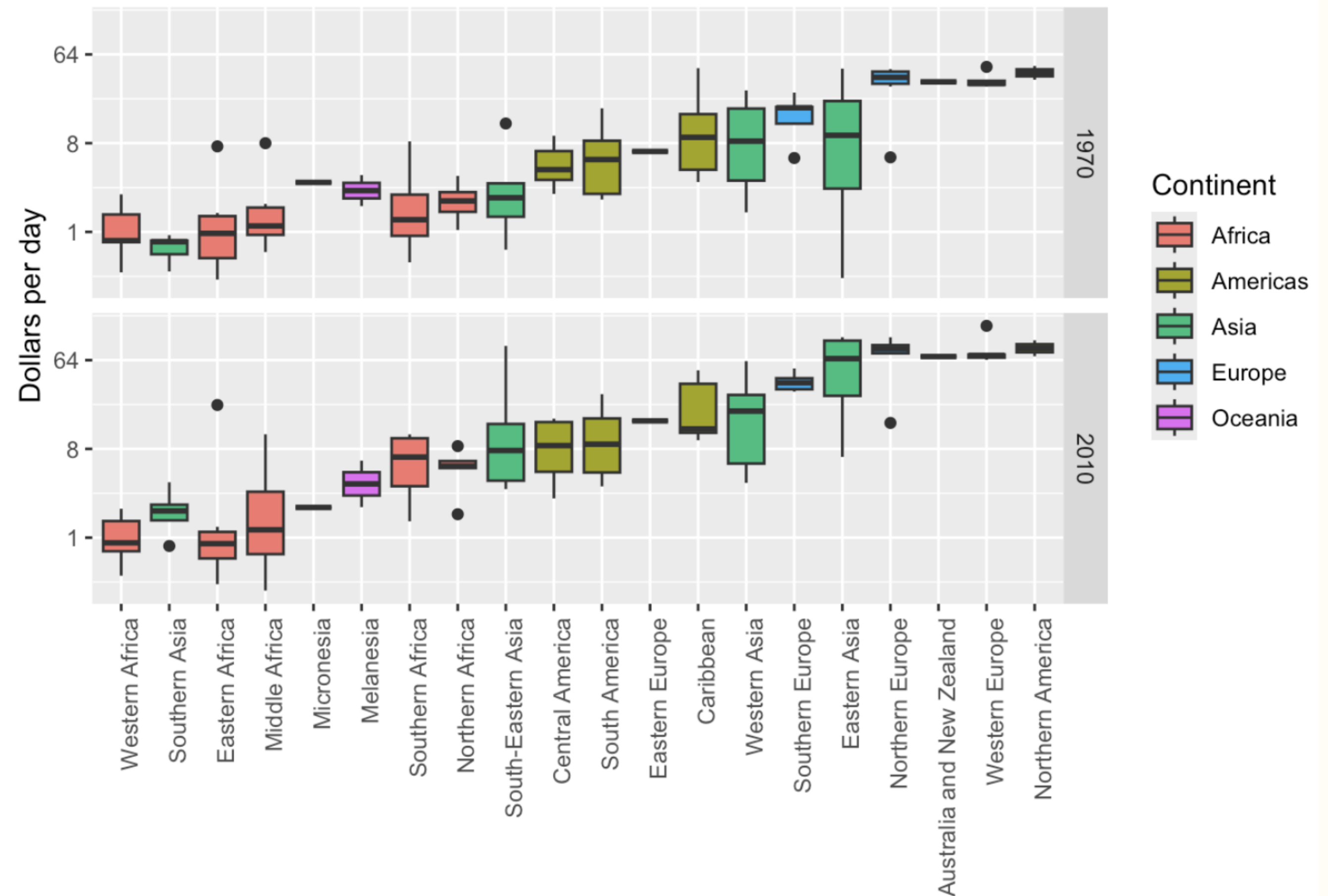
Make valid comparisons

```
gapminder %>%
  filter(year %in% c(past_year, present_year) &
         country %in% country_list) %>%
  mutate(group = ifelse(region %in% west, "West", "Developing")) %>%
  ggplot(aes(dollars_per_day)) +
  geom_histogram(binwidth = 1, color = "black") +
  scale_x_continuous(trans = "log2") +
  facet_grid(year ~ group) +
  xlab("Number of countries") +
  ylab("Dollars per day")
```



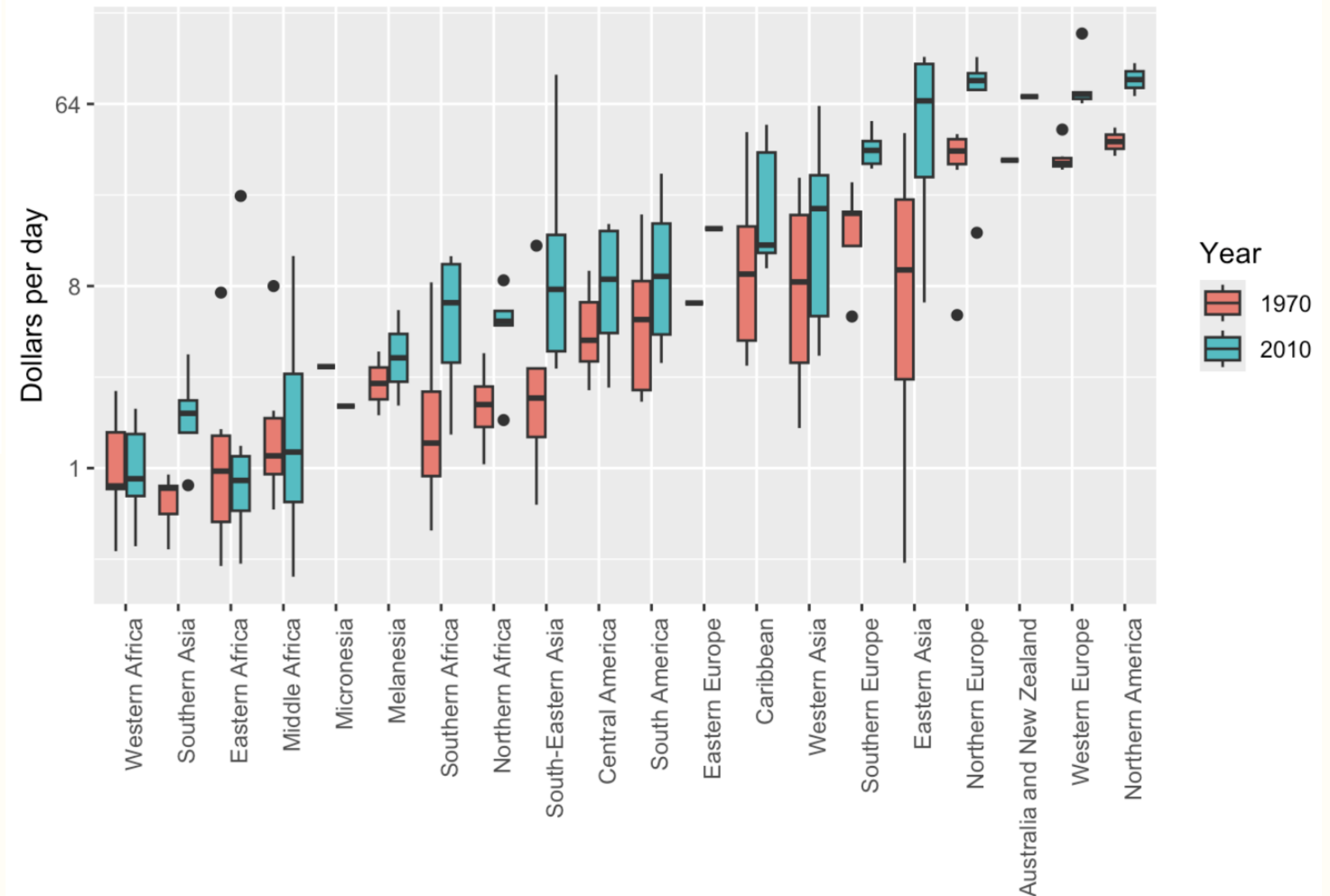
Boxplots across years

```
gapminder %>%
  filter(year %in% c(past_year, present_year) &
         country %in% country_list) %>%
  mutate(region = reorder(region, dollars_per_day, FUN = median)) %>%
  ggplot(aes(region, dollars_per_day, fill = continent)) +
  geom_boxplot() +
  theme(axis.text.x = element_text(angle = 90, hjust = 1)) +
  xlab("") +
  ylab("Dollars per day") +
  scale_y_continuous(trans = "log2") +
  labs(fill = "Continent") +
  facet_grid(year~.)
```



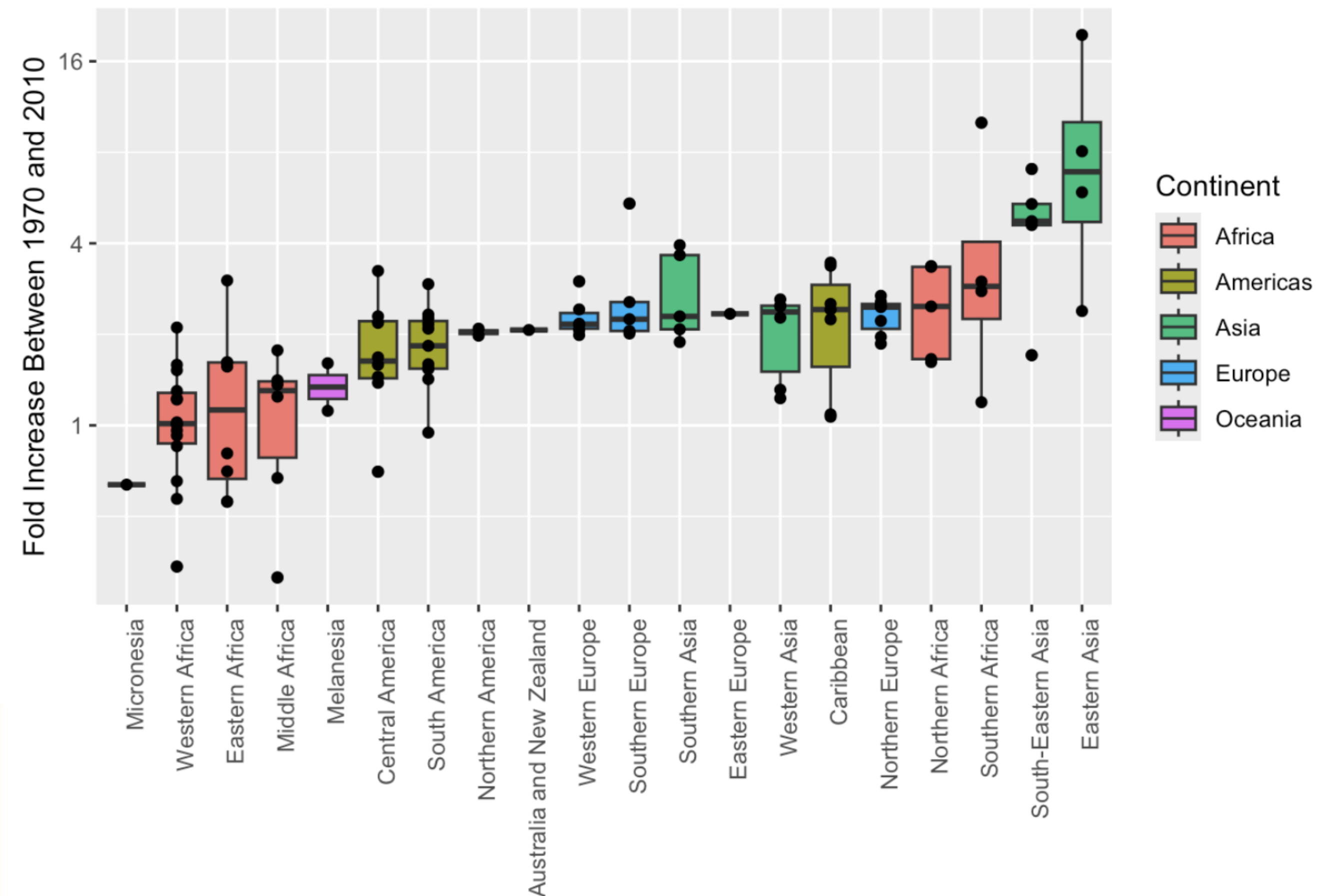
Plot side-by-side

```
gapminder %>%  
  filter(year %in% c(past_year, present_year) &  
         country %in% country_list) %>%  
  mutate(region = reorder(region, dollars_per_day, FUN = median)) %>%  
  ggplot(aes(region, dollars_per_day, fill = factor(year))) +  
  geom_boxplot() +  
  theme(axis.text.x = element_text(angle = 90, hjust = 1)) +  
  xlab("") +  
  ylab("Dollars per day") +  
  scale_y_continuous(trans = "log2") +  
  labs(fill = "Year")
```



Fold change

```
gapminder %>%
  filter(year %in% c(past_year, present_year) &
         country %in% country_list) %>%
  mutate(year = ifelse(year == past_year, "past", "present")) %>%
  select(country, region, continent, year, dollars_per_day) %>%
  spread(year, dollars_per_day) %>%
  mutate(difference = present/past) %>%
  mutate(region = reorder(region, difference, FUN = median)) %>%
  ggplot(aes(region, difference, fill = factor(continent))) +
  geom_boxplot() +
  geom_point(show.legend = FALSE) +
  theme(axis.text.x = element_text(angle = 90, hjust = 1)) +
  xlab("") +
  ylab("Fold Increase Between 1970 and 2010") +
  labs(fill = "Continent") +
  scale_y_continuous(trans = "log2")
```



Summary

- Reordering by a meaningful value eases interpretation
- Comparisons of groups over time are only valid if the groups remain the same over time
- When comparing boxplots, plotting side-by-side rather than using faceting eases interpretation